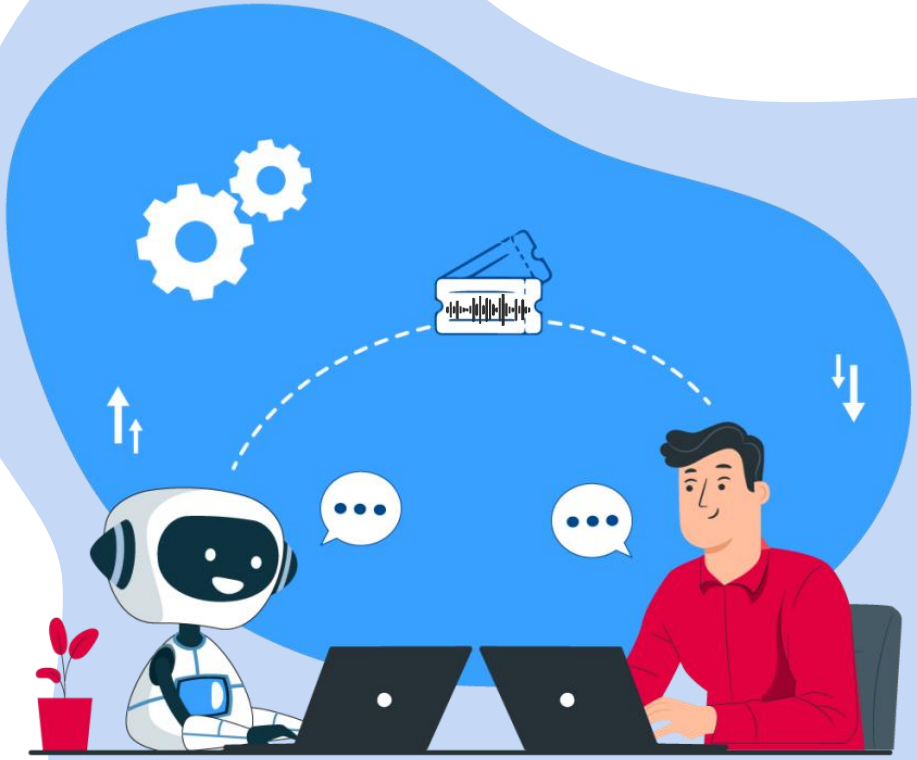


SpeakFree

AI Online Speech Disorder Screening & Therapy Assistant

R26-IT-086

Members

SpeakFree

Supervisor
Ms. Chathurangika Kahandawaarachchi

Co-supervisor
Ms. Suranjini Silva



Group Leader
Kusumsiri P.A.S.S



Tharushan I.L.S



Randiw E.Y



Perera M.D.I

SLIIT FACULTY OF COMPUTING

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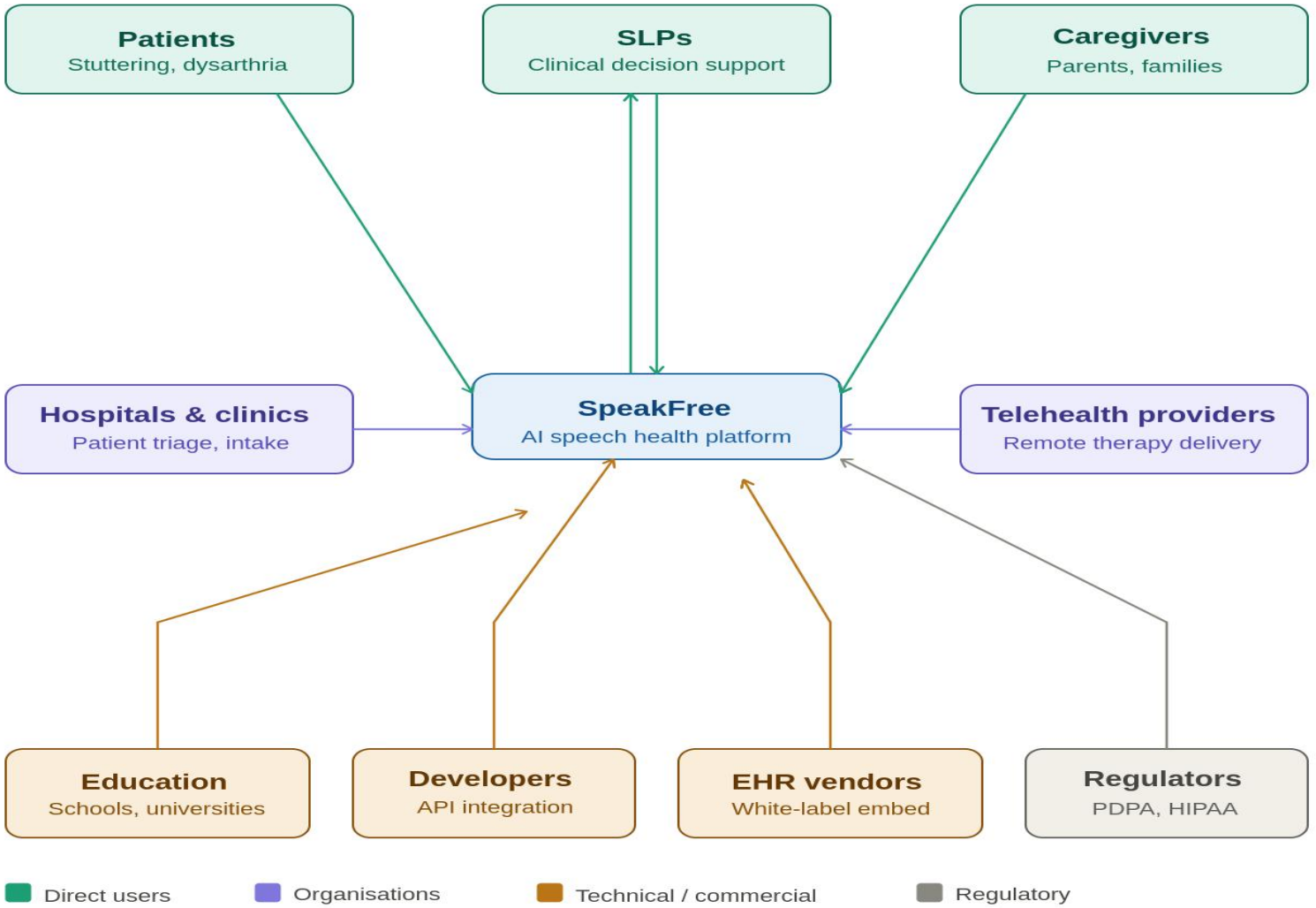
Commercialization Aspects

Budget

Introduction

An online platform that screens speech from a simple browser recording, explains what it finds at the phoneme level, predicts severity and guides personalised home therapy, so quality speech care does not depend on where you live.

Users and Stakeholders



Exiting Solutions

Feature	Stammurai	SpeechViewer	SALT	Constant Therapy	
Multi-disorder simultaneous detection	✗	✗	✗	✗	✓
Real-world noise robustness	✗	✗	✗	✗	✓
Clinical severity grading (Level 1–4)	✗	✗	✗	✗	✓
Real-time adaptive therapy	✓	✗	✗	✓	✓
Formal privacy guarantee (DP + FL)	✗	✗	✗	✗	✓
Cost per session	USD 150–300	USD 150–300	USD 150–300	USD 50–100	**USD 2–5** 97% cost reduction

SpeakFree — R26-IT-086 | Existing Solutions vs. Proposed Solution

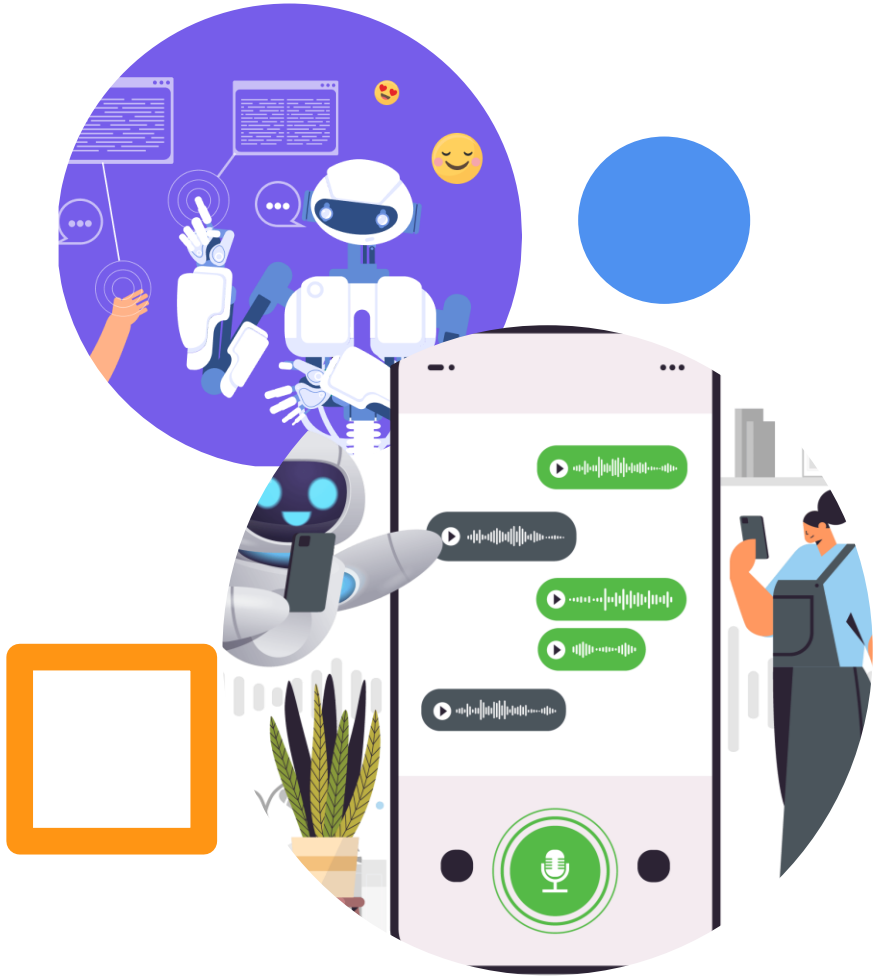
Proposed Solution - SpeakFree

A privacy-preserving, browser accessible mobile application that screens for Stuttering, Dysarthria in primary school students using noise robust feature extraction, phoneme level articulation analysis, multi-disorder severity classification (Level 1 to 4), and an AI adaptive therapy assistant all without centralizing sensitive voice data.

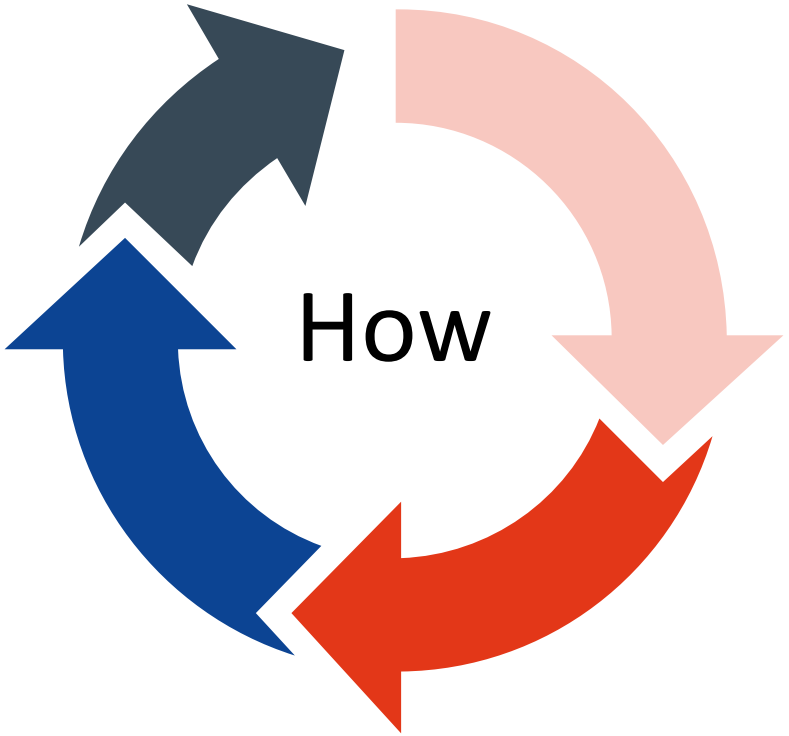
Main Objectives

Main Objective

- To develop an AI powered online platform that provides affordable speech disorder screening and real-time therapy support for people who cannot easily access speech therapists, while ensuring strong privacy protection.



RESEARCH QUESTION



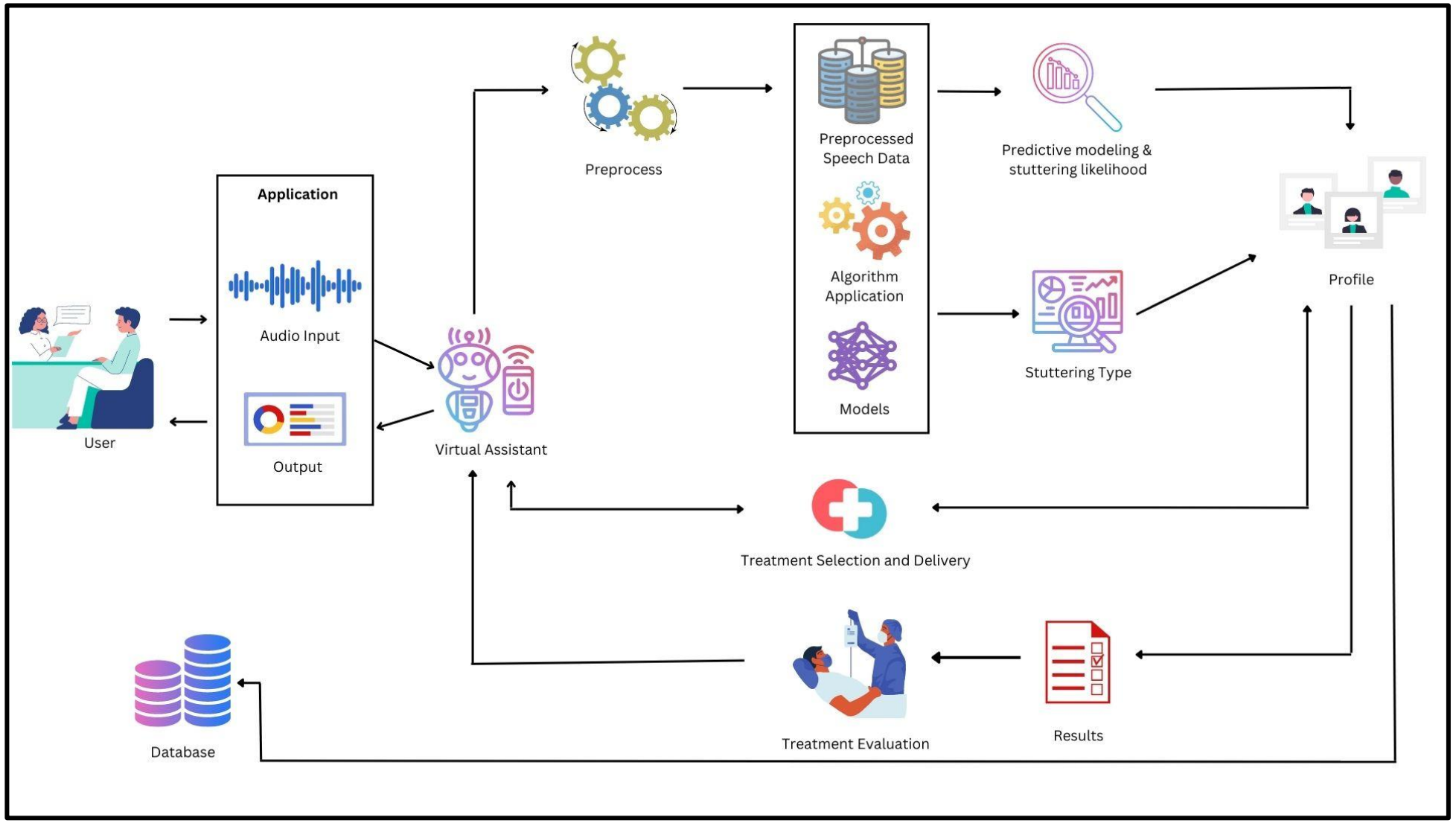
- can speech features be extracted accurately in noisy online environments?
- can articulation errors be detected at phoneme level automatically?
- can AI classify speech disorder severity reliably?
- can real-time speech therapy be delivered while protecting privacy?



Specific Objectives

- Develop a noise-robust speech feature extraction module.
- Develop an articulation analysis module for phoneme level accuracy.
- Develop an AI model for severity classification.
- Develop a privacy-preserving real-time therapy module.

System Overview Diagram



Individual Contributions

P.A.S.S. Kusumsiri **IT22132628**	**Component 1 - Feature Extraction** Builds a browser-based speech capture system that records patient audio and removes background noise. Uses fine-tuned **HuBERT** and **Whisper** models to extract speech fluency features and generate clean speech data for the next system components.
E.Y. Randiw **IT22257086**	**Component 2 - Articulation Analytics** Uses **Montreal Forced Aligner** to identify speech sound boundaries and measure pronunciation accuracy. Produces clear sub-scores such as phoneme timing differences, articulation accuracy, and confidence levels for the severity analysis component.
I.L.S. Tharushan **IT21315114**	**Component 3 - Severity Classification** Builds an AI model that detects multiple speech disorders at the same time, including stuttering and articulation severity, and classifies them into a Level 1 - 4 severity scale. It also adds confidence checking so predictions are reliable and safe for clinical use.
M.D.I. Perera **IT22056702**	**Component 4 - AI Therapy & Privacy** Develops a real-time AI therapy coach that gives instant speech guidance and personalized exercises during practice sessions. It also protects user privacy by improving the model without sending raw voice recordings outside the patient's device.

Technologies

Application:

- Android (Kotlin / Java)
- Recact.js + WebAudio API
- WebSocket Streaming

Database:

- Database Management Systems (DBMS) such as MySQL, MongoDB
- SQLite (on-device)

Middleware Technologies:

- Python 3.11
- TensorFlow 2.x + PyTorch
- Flask REST API
- OpenCV

Technical Concepts:

- Machine Learning
- Convolution Neural Network
- Image Preprocessing
- Voice Recognition (FFT)
- Self Supervised Learning

Algorithms & Models:

- HuBERT – Large fine-tuning
- Whisper (OpenAI) fine-tuning
- Montreal Forced Aligner (MFA)
- Support Vector Machine (SVM)
- DP-SGD (Differential Privacy)

Team Collaboration

- GitHub
- MS Teams

Data Collection and Ethics

1. **Data Collection & Ethics**
2. 20–40 consented English speech samples collected from voluntary participants
3. Controlled and semi-realistic recording environments
4. Reading + spontaneous speech samples
5. Used for validation, calibration, and usability testing only
6. Combined with public datasets
   - **SEP-28k** (https://www.kaggle.com/datasets/ikrbasak/sep-28k) – for stuttering event detection
   - **TORGO** (https://www.kaggle.com/datasets/pranaykoppula/torgo-audio/data) - for dysarthria severity modeling
   - **UA-Speech** (https://www.kaggle.com/datasets/aryashah2k/noise-reduced-uaspeech-dysarthria-dataset) - for cross-dataset validation

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IT22132628 | Kusumsiri P.A.S.S

BSc (Hons) Degree in Information Technology (specialization in Information Technology)

COMPONENT 01

Noise Robust Online Speech Screening and Self Supervised Feature Extraction

BACKGROUND

Component Contribution

Builds the front-end speech capture and feature extraction pipeline, providing clean speech features and confidence-based fluency metrics for later system components.

Research Novelty

Introduces a new speech screening pipeline that combines ASR confidence scoring, speech quality filtering, and three speech tasks in one browser session for more reliable disorder screening.

RESEARCH PROBLEM & GAP

Standard ASR performs poorly on disordered speech, with error rates up to 60–80%.

Fluency metrics like WPM, articulation rate, and disfluency density become unreliable when transcriptions fail, yet no system handles this.

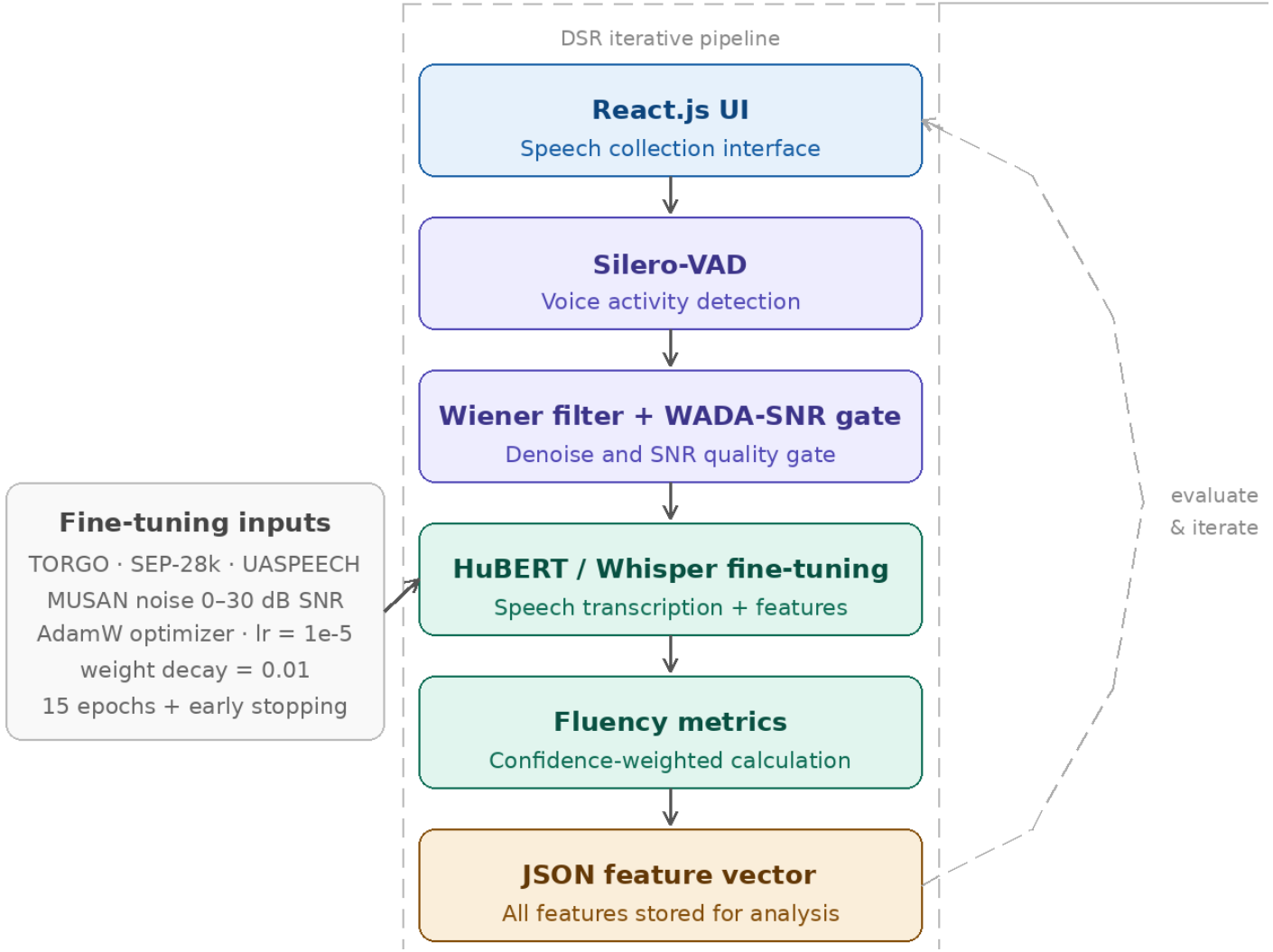
Existing tools are tested only in clean lab conditions, no evaluation exists for noisy, real-world environments.

Single-task read-aloud underestimates stuttering severity in speakers with situationally variable speech.

SYSTEM OBJECTIVES

1. Develop a confidence-aware ASR to handle disordered speech robustly.
2. Generate fluency metrics only from reliable transcriptions to maintain clinical relevance.
3. Design noise-robust feature extraction for realistic, real-world conditions.
4. Implement multi-task elicitation (reading, picture description, spontaneous speech) for accurate assessment of variable speech patterns.

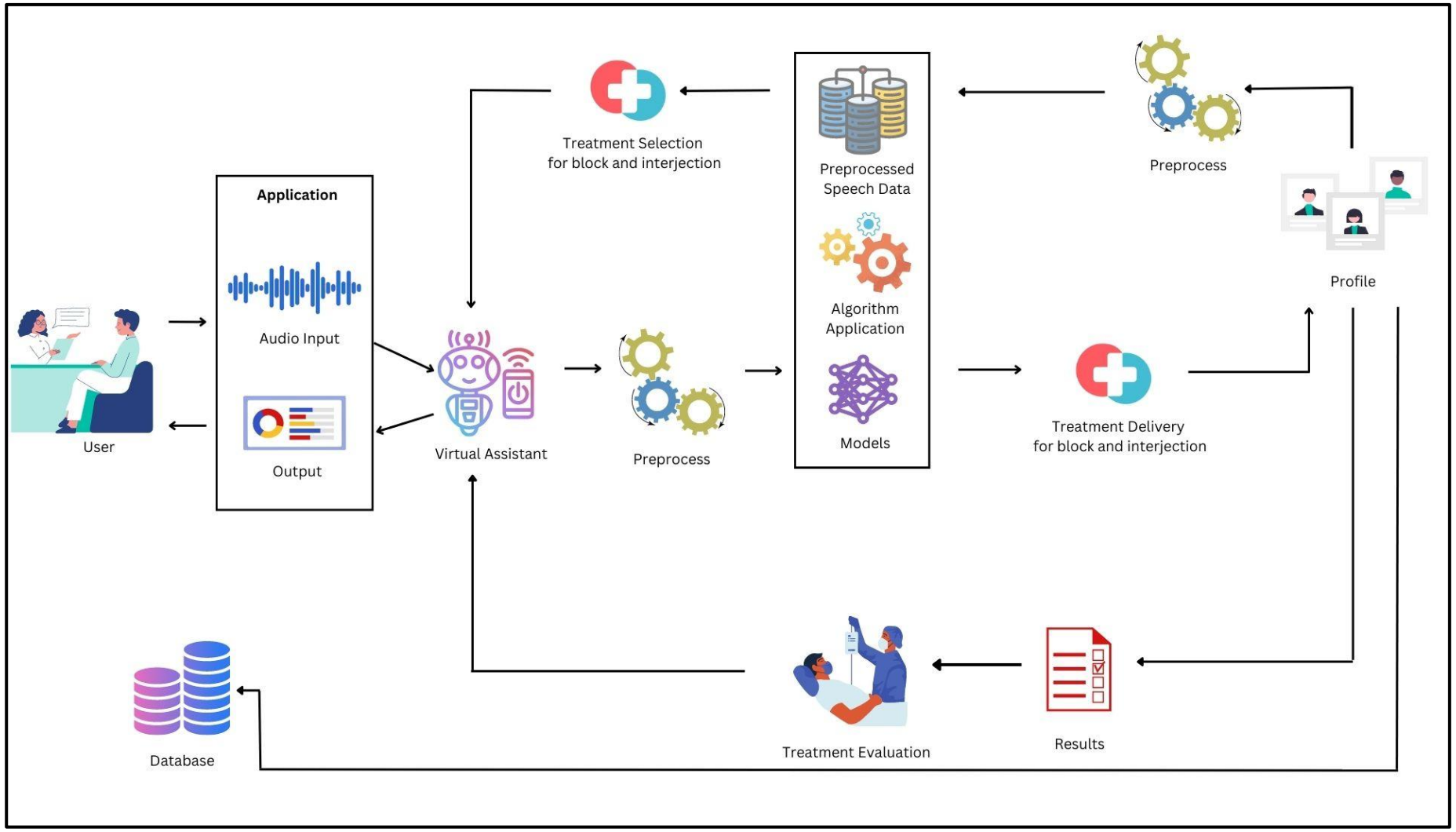
RESEARCH METHODOLOGY & KNOWLEDGE DOMAIN



TECHNOLOGIES AND JUSTIFICATION

Python 3.11 + PyTorch 2.x + HuggingFace Transformers (HuBERT-Large, Whisper-Medium)	• best-in-class pre-trained speech encoders for pathological speech.
Silero-VAD + Librosa + ONNX Runtime	• lightweight ONNX-exportable neural VAD, acoustic features, (<2s inference per 60s audio
React.js + Web Audio API	• browser-native, zero-install, supports live waveform SNR indicator

COMPONET OVERVIEW DIAGRAM



Validation Metrics

- Correlation -Pearson r > 0.85 (clean), r > 0.75 (15 dB SNR)
- Prediction Accuracy - RMSE reduction ≥ 15%, Macro F1 ≥ 0.78
- Speech Detection - VAD F1 ≥ 0.92
- Performance: -Latency < 5 seconds
- User Experience - SUS (System Usability Scale) ≥ 70

Evaluation Methods Cross-Corpus Noise Robustness

- Tested across SNR levels 0–30 dB, three microphone types, and three noise categories.
- ASR Confidence Impact: Compared performance using confidence weighting (θ=0.3) versus unweighted baseline.
- Foundation Model Ablation: Evaluated Whisper vs. HuBERT using leave-one-speaker-out cross-validation.
- Usability Study: Conducted with ~15 SLIIT volunteers, measuring System Usability Scale (SUS).

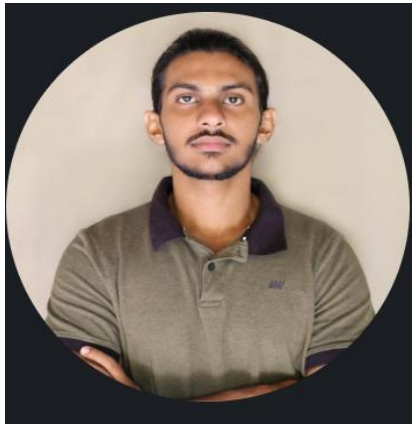
Data Ethics:

Datasets - TORGO, SEP-28k, UASPEECH, and MUSAN – all used under public academic licenses.
Usability - Participants provided written consent, were 18+ years old, and retained the right to withdraw at any time.
Security - No raw audio is stored on the server; session data is anonymized.
Compliance - Fully adheres to PDPA 2022 regulations.

Dependencies & Integration Points

- Component 2 (Randiw) - Receives JSON input containing 768-dimensional embeddings, ASR transcript, phoneme alignment, and confidence-weighted fluency features (WPM, articulation rate, pause frequency, disfluency density, SNR flags).
- Component 3 (Tharushan) - Uses the same JSON vector with ASR confidence-weighted fluency metrics and per-segment reliability indicators for the multi-label classifier.
- Hardware Requirements -Dual-core CPU, 4 GB RAM, and a standard laptop microphone accessible via any modern browser.

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IT22257086 | Randiw E.Y

BSc (Hons) Degree in Information Technology (Specialization in IT)

COMPONENT
02

Alignment and Articulation Analytics

BACKGROUND

Component Contribution

This component analyzes speech at the phoneme level using forced alignment. It measures important speech features such as phoneme duration differences, articulation accuracy, and boundary confidence. These measurements are converted into clear articulation scores, which are then provided to the severity classification system to help determine the level of the speech disorder.

Research Novelty

This component uses phoneme level forced alignment with the Montreal Forced Aligner to analyze speech in detail. It calculates articulation measures such as phoneme duration variation, articulation accuracy, and boundary confidence to evaluate how accurately speech sounds are produced.

Most existing systems use forced alignment tools mainly for linguistic research and offline analysis, rather than in clinical speech assessment systems. This research addresses that gap by integrating forced alignment into a clinic friendly AI application and adding alignment confidence scoring with boundary uncertainty detection. This approach makes speech analysis more reliable and interpretable for automated clinical assessment.

RESEARCH PROBLEM

- Many AI speech analysis systems can recognize words, but they cannot analyze how each sound is actually pronounced.
  Because of this, subtle pronunciation errors related to speech disorders are difficult to detect automatically.



RESEARCH GAP



Most AI speech systems focus on general speech recognition rather than analyzing how each phoneme is pronounced.



Existing phoneme alignment tools are mainly used in research and are rarely integrated into **AI based** speech disorder screening platforms.



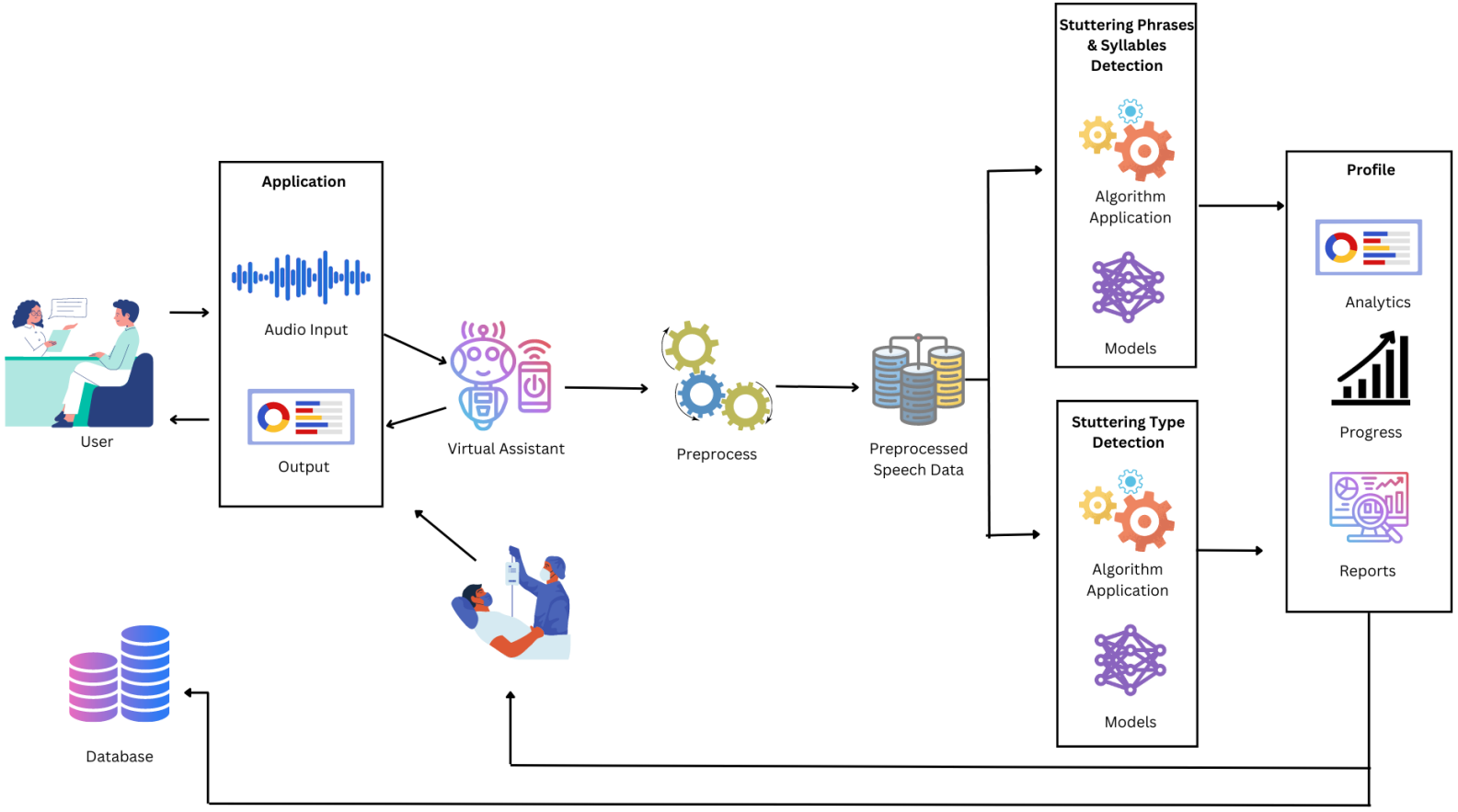
Current systems often provide predictions but do not generate clear phonetic scores that clinicians can easily interpret for assessment.

OBJECTIVES

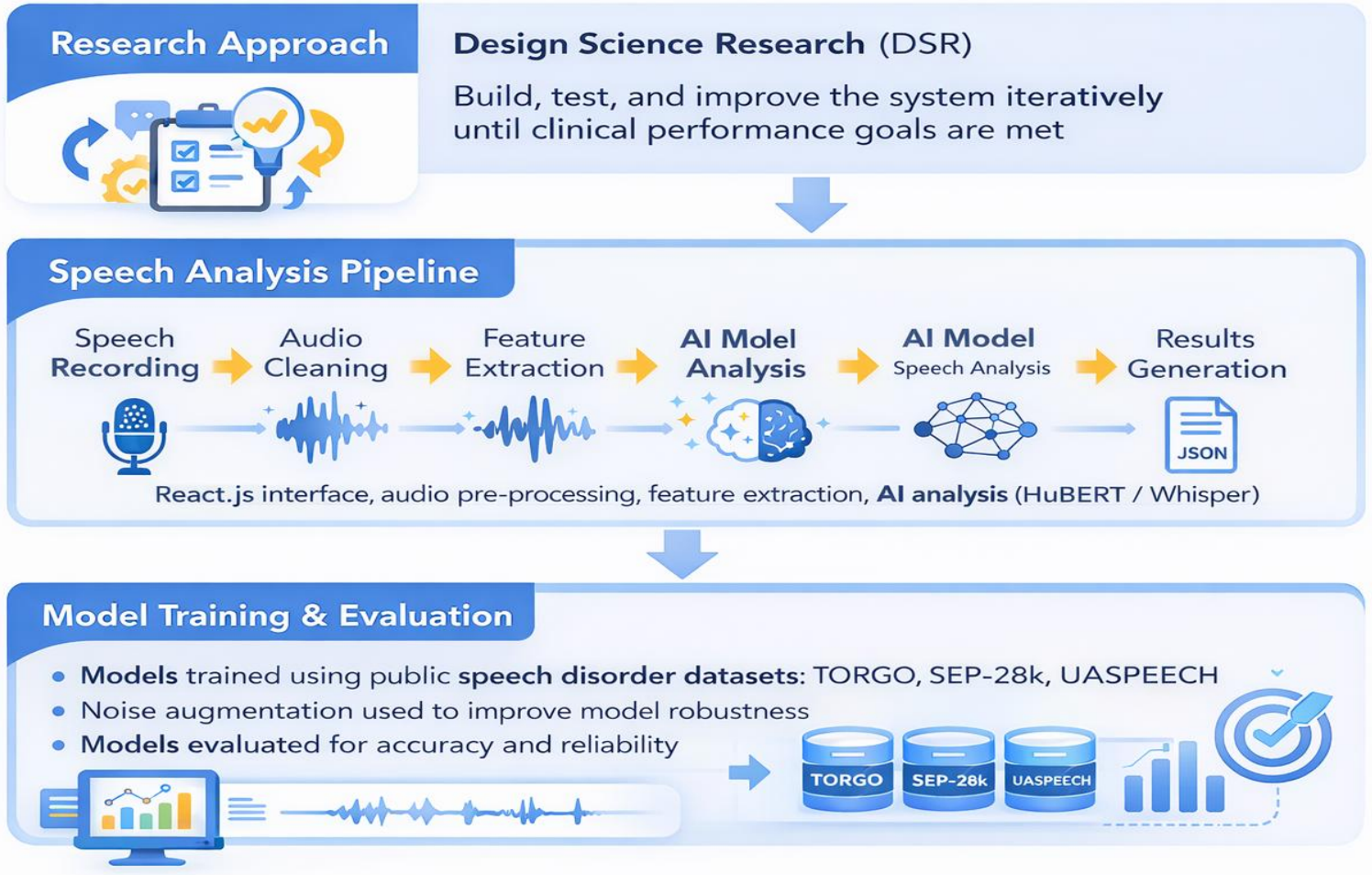
1. Phoneme-Level Speech Alignment - Map speech audio to phonetic transcripts and detect start/end time of each phoneme using Montreal Forced Aligner.
2. Articulation Feature Extraction - Extract phoneme duration, timing variation, pronunciation accuracy, and speech rhythm using Libros and Praat
3. Misarticulation Detection & Phonetic Scoring - Identify incorrect pronunciation patterns and generate per-phoneme articulation accuracy scores
4. Alignment Confidence Estimation - Calculate confidence levels for phoneme boundary detection and flag uncertain segments
5. Structured Output for Classification Module - Generate articulation metrics and send them to the severity classification module through a structured API

METHODOLOGY

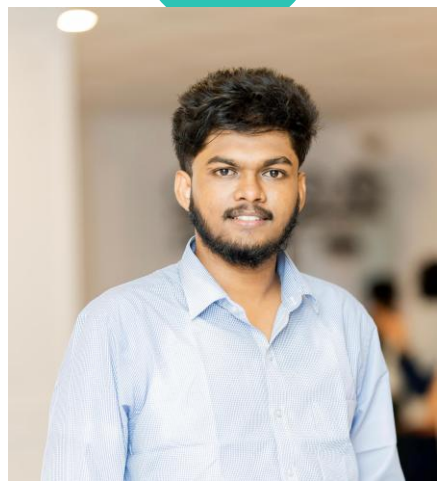
COMPONET OVERVIEW DIAGRAM



RESEARCH METHODOLOGY & KNOWLEDGE DOMAIN



SpeakFree



IT21315114 | Tharushan I.L.S

BSc (Hons) Degree in Information Technology (Specialization in Information Technology)

COMPONENT 03

Multi-Disorder Severity Classification & Risk Prediction

BACKGROUND



❖ Stuttering Events

- **Mono-Syllable Repetition**: Repeating a single syllable within a word or phrase.
  e.g. – "I-I-I want to go" instead of "I want to go"

- **Multi Syllable Repetition**
  Involves a repetition of multiple syllables within a phrase
  e.g. – "mommy- mommy-mommy let's go" instead of "mommy let's go"

❖ Prolongations

Stretching a sound longer than usual.
e.g. – "ssssso" instead of "so"

❖ Blocks

Sudden stoppage or inability to produce sound.
e.g. – Pausing without sound before saying a word

RESEARCH GAP



Existing systems detect only **one disorder at a time**.



ASR-based features are **unreliable for severe speech disorders**.



No unified AI system simultaneously detects and grades multiple speech disorders with reliable uncertainty estimates.

RESEARCH PROBLEM

How to

- Builds an AI model that detects multiple speech disorders at the same time, including stuttering and articulation severity, and classifies them into a Level 1–4 severity scale. It also adds confidence checking so predictions are reliable and safe for clinical use



OBJECTIVES

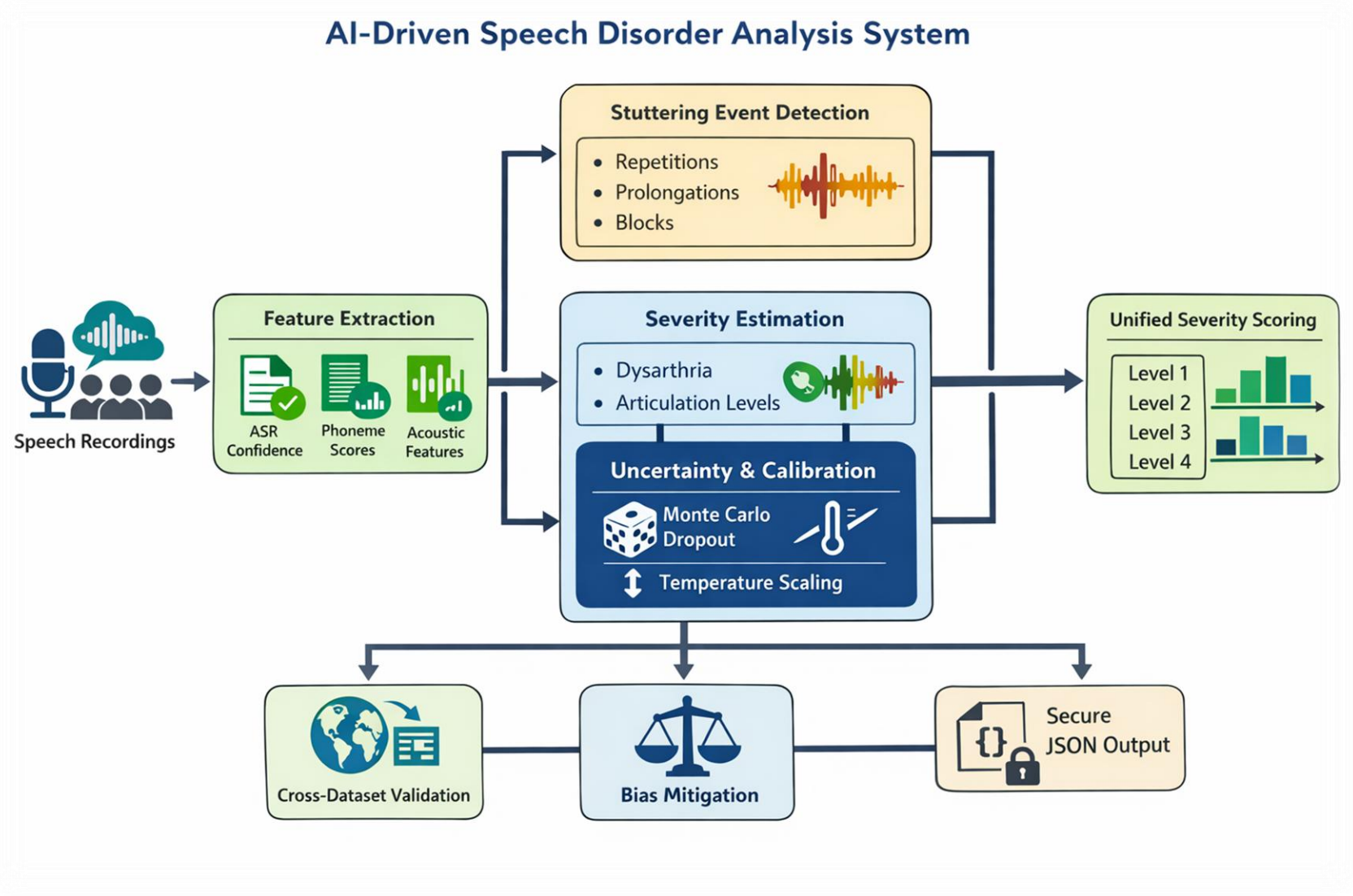
Main Objective

- Develop an **AI system to detect stuttering events and assess speech disorder severity from speech recordings.**

Sub Objectives

- Detect **repetitions, prolongations, and blocks** in speech.
- Predict **dysarthria severity levels** using machine learning.
- Validate the model across **multiple speech datasets**.
- Improve reliability using **uncertainty estimation**.
- Generate **clinical severity scores and structured outputs**.

System Overview Diagram



KEY PILLARS

1. Speech Feature Extraction
2. Multi-Label Stuttering Detection
3. Speech Disorder Severity Estimation
4. Cross-Dataset Validation & Bias Mitigation
5. Uncertainty & Probability Calibration
6. Unified Severity Scoring System

SpeakFree



IT22056702 Perera M.D.I

BSc (Hons) Degree in Information Technology (specialization in Information Technology)

COMPONENT 04

AI Therapy Assistant & Privacy-Preserving Adaptive Learning

BACKGROUND

- **Blocks**

  Complete stoppage of speech or the inability to initiate speech [1].

  e.g. - I want a (pause) pen.

- **Interjections**

  Insertion of different sounds (uh, uhm).

  e.g. - I um need to go home



RESEARCH GAP



Delivery of accurate personalized treatments for blocks and interjections is still a challenge.



More research is needed on discovering more engaging and encouraging techniques of providing treatments for blocks and interjections.



Aims to utilize a new approach of using a virtual assistant to provide treatments for blocks and interjections.

RESEARCH PROBLEM

How to

- Develops a real-time AI therapy coach that gives instant speech guidance and personalized exercises during practice sessions. It also protects user privacy by improving the model without sending raw voice recordings outside the patient's device.



OBJECTIVES

Main Objective

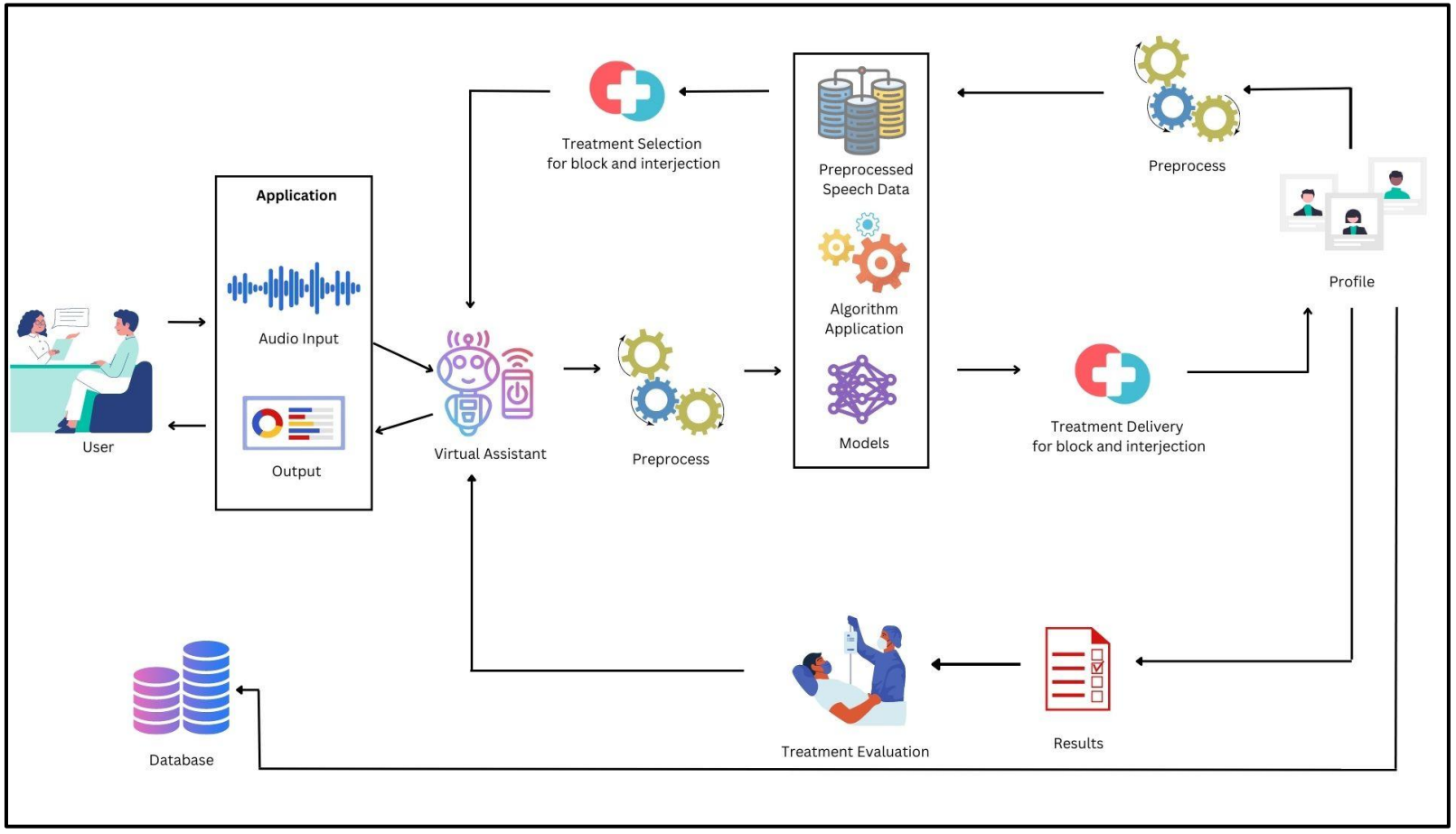
- Provide personalized treatments for patients suffering **from block and interjection** stuttering types.

Sub Objectives

- Identify treatments based on the patient's data.
- Provide treatments using the virtual assistant.
- Evaluate the progress of the patient.

METHODOLOGY

COMPONENT OVERVIEW DIAGRAM



Requirements



Domain expertise in speech-language pathology and stuttering is essential.



Use a chatbot platform like Dialogflow, Microsoft Bot Framework, or IBM Watson Assistant.



NLP is critical for understanding and responding to user queries.



A comprehensive knowledge base is needed, covering stuttering causes, symptoms, and treatment options.



Speech recognition technology is needed to transcribe speech inputs accurately.

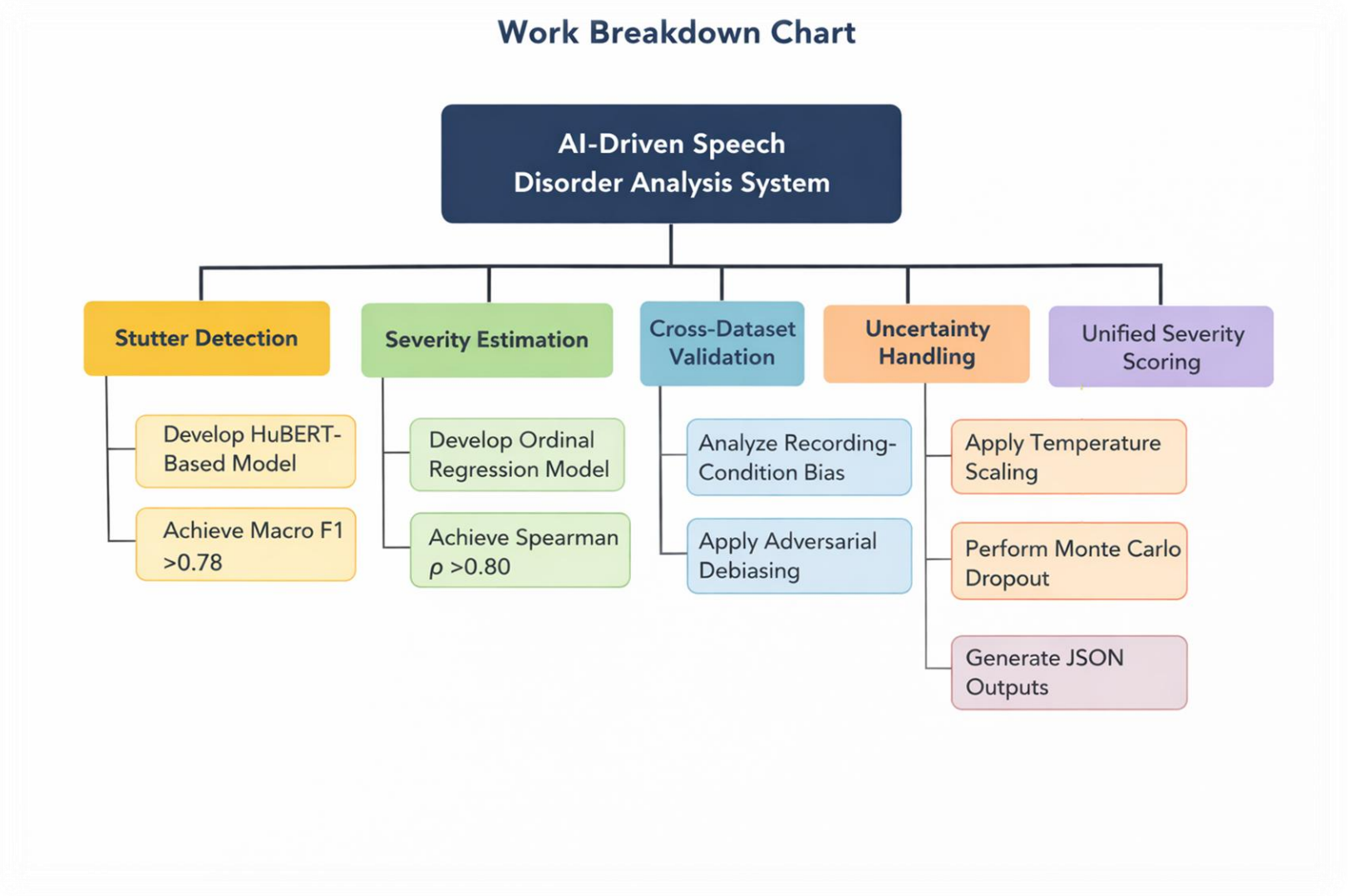


The chatbot's user interface should be designed to provide a seamless user experience.

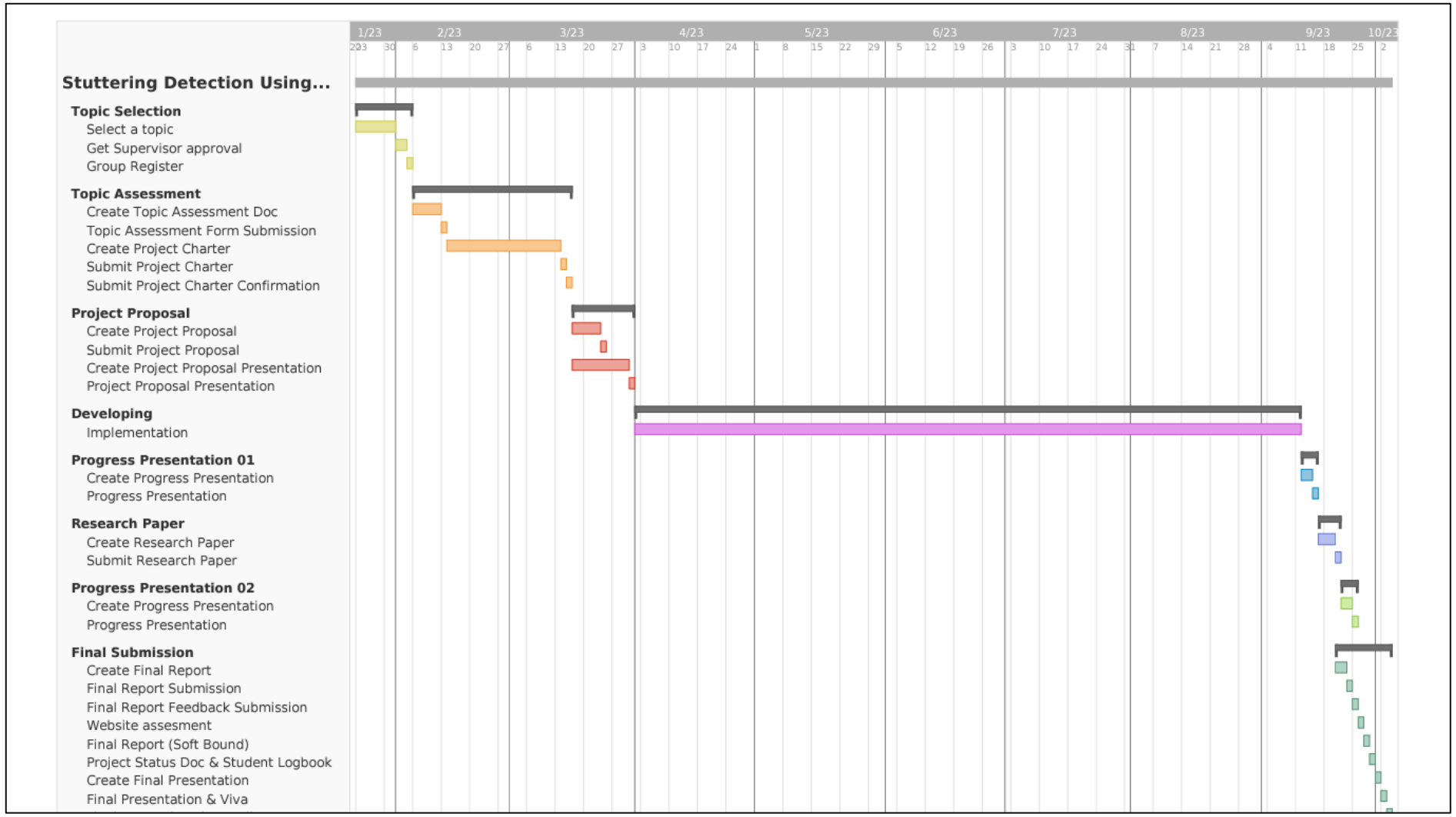


Continuous improvement is needed, including user feedback and ongoing research into stuttering detection and treatment.

Work Breakdown Plan



Gantt Chart


1/23
2/23
3/23
4/23
5/23
6/23
7/23
8/23
9/23
10/23
Stuttering Detection Using...
Topic Selection
Select a topic
Get Supervisor approval
Group Register
Topic Assessment
Create Topic Assessment Doc
Topic Assessment Form Submission
Create Project Charter
Submit Project Charter
Submit Project Charter Confirmation
Project Proposal
Create Project Proposal
Submit Project Proposal
Create Project Proposal Presentation
Project Proposal Presentation
Developing
Implementation
Progress Presentation 01
Create Progress Presentation
Progress Presentation
Research Paper
Create Research Paper
Submit Research Paper
Progress Presentation 02
Create Progress Presentation
Progress Presentation
Final Submission
Create Final Report
Final Report Submission
Final Report Feedback Submission
Website assesment
Final Report (Soft Bound)
Project Status Doc & Student Logbook
Create Final Presentation
Final Presentation & Viva

Commercialization Aspects



Permanent Subscription
Rs. 3000



Monthly Subscription
Rs. 500



Annual Subscription
Rs. 2000



Government Speech Clinics Subscription
Rs. 800



Children Orphanages
Free



Social Media Commercialization.

Budget

Component	Est. Amount (USD)	Est. Amount (LKR)
Internet connectivity charges	9.4	3000
Charges for tools used for the research (Cloud Services, Grammarly, Mendeley etc)	31.37	10000
Travelling fees for consultation sessions and data collecting (Medcon (pvt) ltd)	2.15	700
Technical consultation charges (External technical information session)	13.9	4500
Cloud Platforms(AWS, GCP, Dialogflow, Microsoft Bot Framework, Amazon Lex)	40	13000

Q & A

Thank You



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